LEARNER GUIDE

Finding an angle

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

So far the angle was given and you found a side. This lesson runs the idea backwards: you have two sides and you want the angle. The method is the same four lines, with one new tool — the inverse ratios $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$, which undo sin, cos and tan and turn a ratio back into its angle. You write the ratio the two sides make, put in the numbers to get a value, then apply the inverse to reveal the angle: for example $\tan\theta = 3 \div 4 = 0.75$, so $\theta = \tan^{-1}(0.75) = 36.9°$. You will use the calculator's $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ keys (in degrees), and read real angles — the steepness of a ramp, the pitch of a roof, an angle of elevation — from two lengths you can measure. The question you are exploring is: how can two sides reveal the angle?

What you are learning to notice

- That two sides are enough to find the angle, using the ratio those two sides make.
- That $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ are the inverse ratios — they undo sin, cos and tan, turning a ratio back into its angle.
- That the angle is $\text{ratio}^{-1}(\text{value})$, never the ratio value itself.
- That finding an angle is finding a side run backwards: choose the ratio, substitute, then apply the inverse instead of rearranging for a side.

Moving through it, one step at a time

In From ratio back to angle, choose the ratio the two sides make, then reveal the working one line at a time — watch the inverse turn the ratio into the angle. In Angle Finder, do it yourself: choose the ratio from the two sides, then type the inverse line $\theta = \text{ratio}^{-1}(\text{value})$, then work out the angle on the calculator to 1 decimal place. In Spot the slip, catch the two classic errors — the wrong ratio, and forgetting the inverse (giving the ratio value as the angle). In Reveal the angle, find a real angle (a ramp, a roof, an angle of elevation) from two measured lengths. Keep the calculator in degrees; its $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ keys are the ones you need.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (SOH · CAH · TOA and choosing a ratio from two sides (Lesson 4), finding a side by writing and solving the ratio equation (Lessons 5–6), and using a calculator's inverse trig keys in degrees), open a short recap and return whenever you are ready.

- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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